

CSCE 748 COMPUTATIONAL PHOTOGRAPHY · MODULE 05 · LECTURE NOTES

Demosaicing and the Raw Pipeline

Two-thirds of the colour data was never measured.

MODULE OUTCOMES

- Explain the colour filter array and why it exists.
- Implement demosaicing and explain its failure modes.
- Explain white balance and the illuminant estimation problem.
- Trace the full raw-to-display pipeline and justify the ordering.
- Explain colour spaces and the camera colour matrix.

1 The colour filter array

A SENSOR MEASURES INTENSITY, NOT COLOUR

A photosite counts photons. It has no mechanism for distinguishing a red photon from a blue one — both liberate an electron, and the well counts electrons.

So each photosite is covered by a colour filter, passing one band and blocking the others. The **Bayer pattern** — a 2×2 tile of one red, one blue, and two greens — is near universal.

Twice as many green sites because human luminance sensitivity peaks in green, and spatial detail is perceived largely through the luminance channel. Sampling green more densely therefore buys more perceived resolution than sampling red or blue would.

The consequence is that two-thirds of the colour information at every pixel is interpolated rather than measured. Demosaicing is that reconstruction, and its errors appear as colour artefacts that no amount of later processing removes.

Method	Approach	Assessment
Nearest neighbour	Copy the value from the nearest photosite of the required colour.	Poor. Blocky, with obvious colour artefacts. Fast, and used only for previews.
Bilinear	Average the surrounding photosites of each colour independently.	Soft, with coloured fringes at edges — because averaging across an edge mixes two different surfaces, and does so differently in each channel.
Edge-directed	Estimate the local edge direction and interpolate <i>along</i> it rather than across it.	Substantially better. Removes most fringing on straight edges.

Method	Approach	Assessment
Gradient-corrected / Malvar	Use the densely sampled green channel's detail to guide the interpolation of red and blue, on the assumption that the channels share structure.	The classic good method. A fixed linear filter, cheap, and close to the quality of much more expensive approaches.
Learned	A network trained on pairs of raw mosaics and ground-truth full-colour images.	Best measured quality, and it invents plausible detail where the measurement is ambiguous — Module 13's question.

The guiding insight is that green is the detail channel. It is sampled at twice the density of the others, so it has the highest-quality structural information, and the chromatic channels can be reconstructed by assuming that colour varies more slowly than luminance — which is true for most real surfaces.

DEMOSAICING ERRORS ARE STRUCTURAL

Zippering: alternating light and dark or coloured pixels along a horizontal or vertical edge, produced by interpolating across the edge rather than along it. The name describes the appearance exactly.

Colour moiré: fine repeating detail near the sampling limit aliases *differently in each channel*, because the channels are sampled on different lattices. The result is coloured banding where the scene is grey — fabric and distant architecture are the classic subjects.

False colour at sharp edges and on fine texture, where the assumption that chrominance varies slowly fails.

All of these are aliasing artefacts in the sense of Module 02 — scene detail above the sampling rate folding down. **This is why cameras have historically included an optical low-pass filter** in front of the sensor, and why removing it is a deliberate trade of sharpness against exactly these artefacts.

2 White balance

COLOUR CONSTANCY IS AN UNDERDETERMINED INFERENCE

The sensor measures reflected light, which is the product of the illumination spectrum and the surface reflectance spectrum. Recovering both from the product is underdetermined by construction.

A white surface under orange light and an orange surface under white light produce identical measurements. There is no information in the pixel that distinguishes them — the ambiguity is fundamental, not a limitation of the sensor.

Humans perform this separation remarkably well, using scene context, memory of what objects are usually coloured, the range of colours present, specular highlights (which carry the illuminant's colour directly), and probably several mechanisms not yet understood. A white shirt looks white indoors and outdoors.

Cameras estimate it with heuristics, each of which encodes an assumption about scenes — and each therefore fails on the scenes where its assumption does not hold.

Heuristic	Assumption	Fails when
Grey world	The average reflectance of a scene is achromatic, so the average of the image reveals the illuminant.	A large uniform colour dominates the frame. A photograph of a forest is estimated as having green light, and the correction turns the foliage grey.
White patch / max-RGB	The brightest pixel in each channel corresponds to a white surface.	There is no white object in the scene, or the brightest pixel is a clipped highlight carrying no colour information. A snow scene has the opposite problem — everything is bright.
Grey edge	The average of the <i>derivative</i> of the image is achromatic.	More robust than grey world in practice, and it shares the same failure in principle.
Gamut mapping	The set of colours observable under a given illuminant is constrained, so the observed gamut narrows the possibilities.	Scenes containing few distinct colours, which constrain little.
Learned	Statistics of illuminants and scenes, from labelled training data.	Scenes unlike the training distribution — which is the standard failure mode of learned components and is covered in Module 12.

3 The pipeline

1. SUBTRACT BLACK LEVEL	the sensor's zero is not 0
2. SCALE to [0,1]	using the saturation value
3. WHITE BALANCE	per-channel multipliers <-- before demosaic
4. DEMOSAIC	one channel -> three
5. COLOUR MATRIX	camera RGB -> XYZ -> working space
6. DENOISE / SHARPEN	in LINEAR space (Module 01)
7. TONE MAP	HDR -> display range (Module 04)
8. GAMUT MAP	handle out-of-gamut colours
9. ENCODE	display transfer function <-- LAST, ONCE

Steps 1 through 6 operate on linear data, which is what makes them physically meaningful (Module 01). **Step 9 is where linearity ends**, and it happens exactly once, at the end.

WHY WHITE BALANCE PRECEDES DEMOSAICING

This ordering surprises people and the reason is specific.

Good demosaicing interpolates *between* channels. The gradient-corrected methods of §1 use the densely sampled green channel's structure to guide the reconstruction of red and blue, on the assumption that the channels share edges.

That assumption requires the channels to be on comparable scales. Before white balance they are not — under tungsten illumination the raw blue channel may be a quarter of the raw red, so a green-guided correction applied to blue is scaled wrongly.

So interpolation guided by uncorrected channels introduces colour errors, and it does so at exactly the edges where demosaicing is already most difficult. **Applying the per-channel multipliers first** brings the channels into comparable ranges before the interpolation depends on it, and the artefacts largely disappear.

- **Camera RGB is device-specific.** It depends on the spectral transmission of that camera's particular filters, and a value in it means nothing outside that camera model.
- **The colour matrix maps camera RGB to CIE XYZ**, a device-independent space defined by the standard observer's cone responses. This matrix is measured per camera model and is stored in the raw file's metadata.
- **Then XYZ to a working space** — sRGB for display, Adobe RGB or ProPhoto for editing, ACEScsg for film pipelines. **Wider spaces preserve saturated colours that sRGB clips**, and they need more bits to avoid banding, which is why wide-gamut work is done in 16-bit or floating point.
- **The mapping is approximate.** A camera's filter responses are not a linear transformation of the human cone responses — the Luther–Ives condition is not satisfied by any real camera — so **no 3×3 matrix is exact**. The fit is optimised over a set of representative colours and is wrong elsewhere.
- **This is why two cameras render the same scene differently** even with identical white balance and identical processing. It is a measurable physical difference in the filters, not a matter of manufacturer preference — though the tone curves are that as well.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
CMU 15-463 — the image processing pipeline lecture and assignment (free)	The complete raw pipeline, implemented. This assignment is Project 1's first requirement.
Malvar, He & Cutler — High-quality linear interpolation for demosaicing (free)	The gradient-corrected method of §1. A fixed linear filter, a few lines, and close to state of the art for a long time.
Gijssenij, Gevers & van de Weijer — Computational Color Constancy survey (free)	The illuminant estimation methods of §2, compared on standard datasets.
Adobe DNG specification (free)	What is actually in a raw file, including the colour matrices and how they are intended to be applied.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Extract the raw Bayer array from a file and visualise it directly. Confirm the pattern and the channel ratios.
- 2 Implement bilinear demosaicing and photograph a sharp high-contrast edge to produce visible fringing.
- 3 Implement the Malvar gradient-corrected method and compare on the same image.
- 4 Photograph fine repeating detail — fabric, a distant fence — and reproduce colour moiré.
- 5 **Demosaic before and after white balance** on a tungsten-lit scene and compare the colour errors at edges.
- 6 Implement grey world and white patch illuminant estimation. Find a photograph that defeats each.
- 7 Implement the full nine-step pipeline and compare your output against the camera's JPEG. Identify the differences and attribute each to a stage.
- 8 Apply the camera's colour matrix and compare against ignoring it. Report the colour error on a chart if you have one.
- 9 Process the same raw file into sRGB and into ProPhoto. Find a colour that one clips and the other does not.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Why does a sensor need a colour filter array, and why is half of the Bayer pattern green?
- 2 What fraction of colour data is interpolated, and what is demosaicing?
- 3 Name five demosaicing methods and the insight the good ones use.
- 4 Name three demosaicing artefacts and say what they are instances of.
- 5 Why is white balance underdetermined? Give the ambiguity precisely.
- 6 Give five illuminant heuristics and a scene that defeats each.
- 7 List the nine pipeline stages in order, and say where linearity ends.
- 8 Why must white balance precede demosaicing?
- 9 Why is the camera colour matrix only approximate, and what does that explain?